

There are three types of I/O devices:



1. **Human readable** – Devices for users to interact with computers, like printers, keyboards, and screens.
2. **Machine readable** – Devices for connecting to electronic equipment, such as disk drives and sensors.
3. **Communication** – Devices for remote connections, like modems.

I/O devices are different in several ways:

1. **Data Rate** – Devices have different speeds for data transfer.
2. **Application** – How a device is used affects the software.
3. **Complexity of Control** – More complex devices need more operating system control.
4. **Unit of Transfer** – Data can be sent as bytes, characters, or larger blocks.
5. **Data Representation** – Devices use different methods to encode data.
6. **Error Conditions** – Devices handle errors in different ways.

Başlıklara bak yeter.

There are **three methods** to perform I/O operations:



1. **Programmed I/O** – The processor sends an I/O command and waits until the operation is done.
2. **Interrupt-driven I/O** – The processor sends an I/O command and continues other tasks if non-blocking. If blocking, it waits for the I/O to finish.
3. **Direct Memory Access (DMA)** – The DMA module handles data transfer between memory and the I/O device directly, without using the processor.



- **I/O-to-memory transfer through processor**
 - **No interrupts:** Programmed I/O
 - **With interrupts:** Interrupt-driven I/O
- **Direct I/O-to-memory transfer**
 - Direct memory access (DMA) is used for direct I/O-to-memory transfer.

I/O Design Goals:



1. **Efficiency**
 - The goal is to speed up I/O because slow I/O can cause delays.
 - I/O devices work much slower than memory and the processor.
 - Disk I/O is the part that needs the most improvement.
2. **Generality**
 - Aim is to manage all devices in the same way.
 - This means processes and the operating system should handle devices similarly.
 - All devices should be handled in a similar way, it is hard to do this.
 - Using a step-by-step and modular design can help manage I/O better.



Buffering helps avoid delays and improve efficiency by handling data transfer before or after a request.

1. Block-oriented device

- Stores data in fixed-size blocks.
- Transfers data one block at a time.
- Data can be accessed by block number.
- Examples: Disks, USB drives.

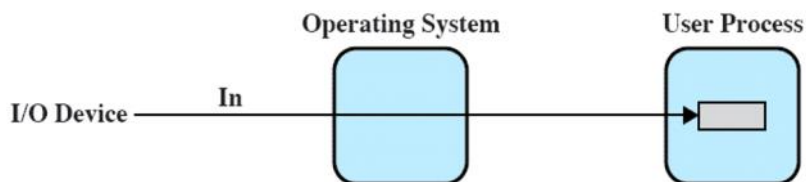


2. Stream-oriented device

- Transfers data as a continuous stream of bytes.
- No block structure.
- Examples: Terminals, printers, communication ports, mouse.

No Buffer

- Without a buffer, the OS directly accesses the device when it needs.

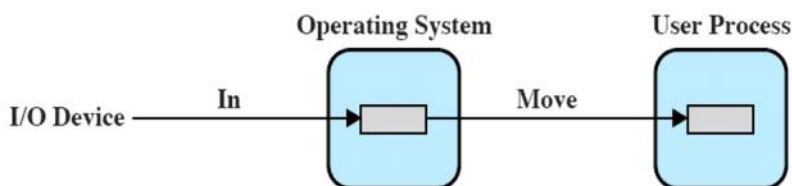


(a) No buffering



Single Buffer

- When user process issues an I/O request. it uses this buffer.



(b) Single buffering



- **Input transfers** go to the system buffer.
- **Reading ahead** is done, expecting that the data will be needed soon.
 - After transfer, the process moves the data and requests more.
- **Speedup** happens because the process works while new data is read.
- **Disadvantages:**
 - Makes OS logic more complex.
 - Swapping becomes harder.



Single Buffering for Stream-Oriented I/O

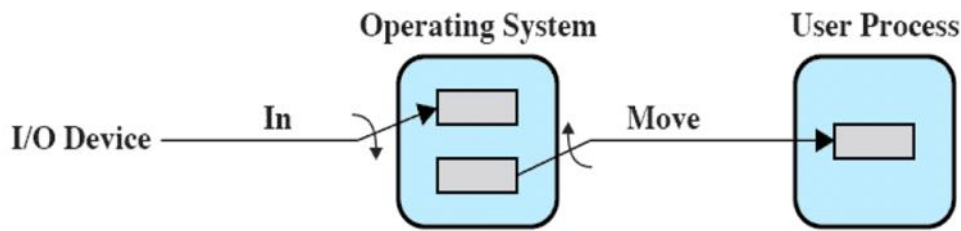
Single buffering handles data one line or one byte at a time.

- **Line-at-a-time:** For simple terminals (one line input/output).
- **Byte-at-a-time:** For devices where each keystroke matters (e.g., sensors, controllers).



Double Buffer

- The process uses one buffer while the OS fills or empties the other.
- This method is called buffer swapping.

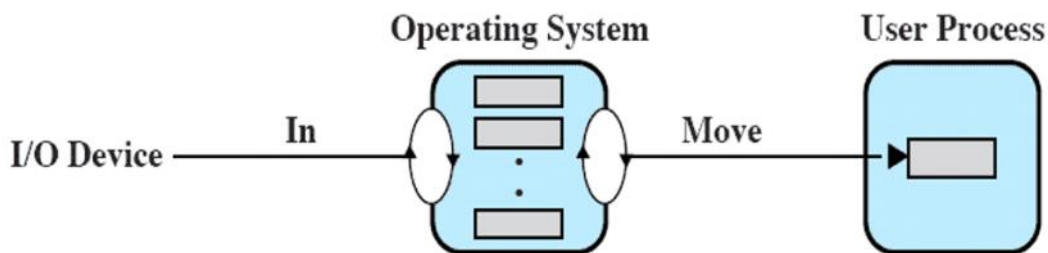


(c) Double buffering

Circular Buffer

Circular Buffer uses more than two buffers.

- All buffers together form a circular buffer.
- Each buffer is a unit in the circular buffer.



(d) Circular buffering

Disk Performance Parameters

1. The disk rotates at a constant speed.
2. To read/write, the head must be at the correct track and sector.
3. **Seek time** is the time to move the head to the right track.
4. **Rotational delay** is the time for the sector to reach the head.
5. **Access time** = Seek time + Rotational delay.



Seek Time

1. Time needed to move the disk arm to the right track.
2. Includes two parts:
 - Startup time
 - Time to cross tracks.
3. **Settling time**: Time to confirm the correct track after positioning.
4. Smaller, lighter parts improve seek time, which is usually under 10ms.



Rotational Delay

Rotational delay means waiting for the disk to spin. The read/write head needs to reach the correct position to access the data.

First-In, First-Out (FIFO)

1. Processes are handled in the order they arrive.
2. It treats all processes equally.
3. If many processes compete, performance may become random.

Priority (PRI)

1. Scheduling is not controlled by disk management software.
2. The goal is to meet other needs, not to improve disk usage.
3. Short and interactive jobs get higher priority.
4. It gives fast response for interactive jobs.
5. Long jobs may wait too long.
6. Not suitable for database systems.